Your notes

# Cambridge (CIE) IGCSE Additional Maths

## Integration

### Contents

Your notes

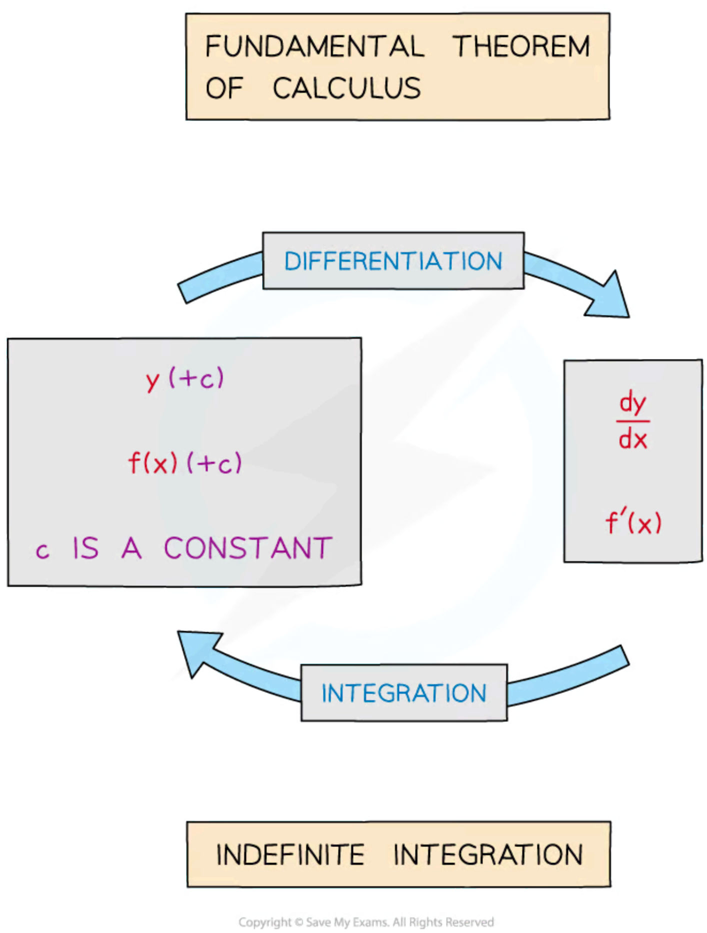
## Introduction to Integration

# Introduction to Integration

## What is integration?

- **Integration** is the opposite to **differentiation**
- **Integration** is the process of finding the expression of a function from an expression of the **derivative** (**gradient function**)

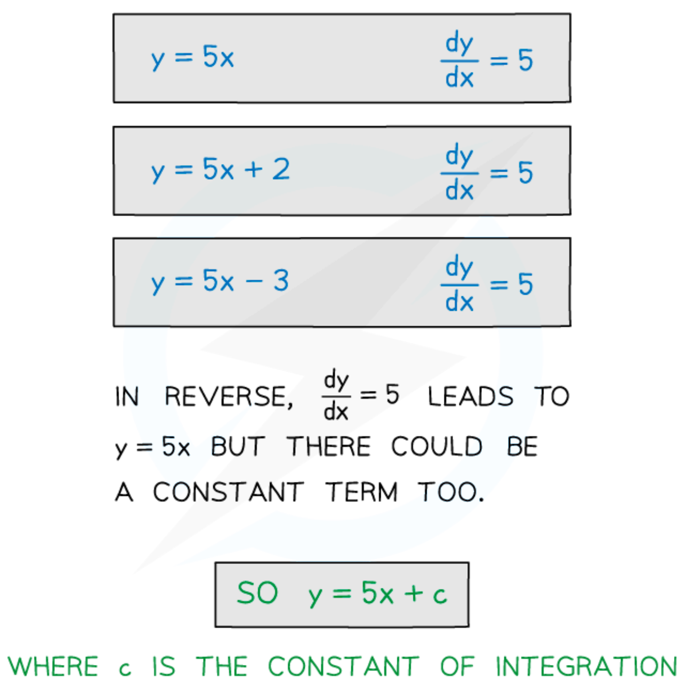
## What is the fundamental theorem of calculus?



- The Fundamental Theorem of Calculus states that integration is the inverse process of differentiation
  - This form of the Theorem relates to **Indefinite Integration**
  - An alternative version of the Fundamental Theorem of Calculus involves **Definite Integration**

## What is the constant of integration (+c)?

- When differentiating ***y***, constant terms 'disappear'
  - for constants ***y* = c**, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$
  - graphs of constants are horizontal lines and so have gradient of 0
- Integrating $\frac{\mathrm{d}y}{\mathrm{d}x}$, to get ***y***, cannot determine the constant
  - To acknowledge this constant, "**+ c**" is used
  - **c** is called the **constant of integration**

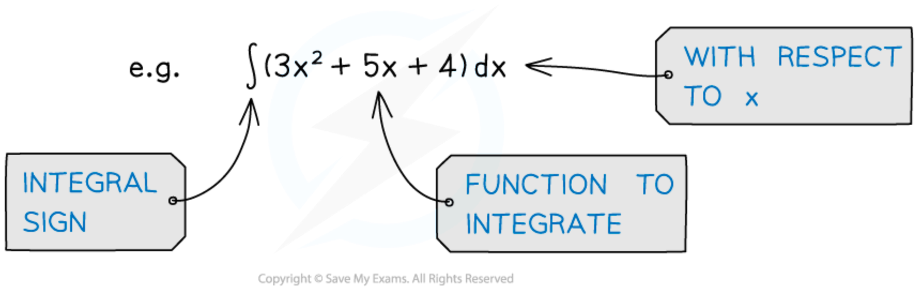


## What is the notation for integration?

- An **integral** is normally written in the form

$$\int f(x)\,\mathrm{d}x$$

- the large operator $\int$ means "integrate"
  - "$\mathrm{d}x$" indicates which variable to integrate with respect to
  - $f(x)$ is the function to be integrated
- If it has more than one term the function to be integrated (called the **integrand**) should be in brackets
  - "Integrate" -- "**all** of (...)" -- "with respect to x"




Your notes

### Worked Example

Your notes

(a) Find $\frac{dy}{dx}$ for $y = 6x + 3$

(b) Find $\int 6 \, dx$

a) $\frac{dy}{dx} = 6$

b) $y = 6x + c$

CONSTANT OF INTEGRATION

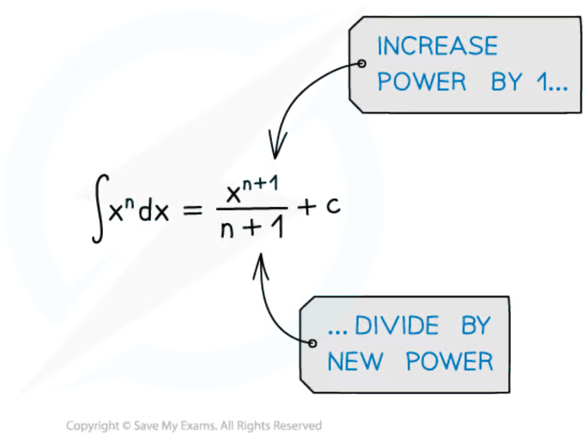
Integrating Powers of x

Your notes

# Integrating Powers of x

## How do I integrate powers of x?

- Powers of $x$ are integrated according to the following formulae:
  - If $f(x) = x^n$ then $\int f(x)\,\mathrm{d}x = \frac{x^{n+1}}{n+1} + c$ where $n \in \mathbb{Q},\ n \neq -1$ and $c$ is the **constant** of **integration**
  - For each term …
    - … increase the power (of x) by 1
    - … divide by the new power
  - This does not apply when the original power is -1
    - the new power would be 0 and division by 0 is undefined




- If the power of $x$ is multiplied by a constant then the integral is also multiplied by that constant
  - If $f(x) = ax^n$ then $\int f(x)\,\mathrm{d}x = \frac{ax^{n+1}}{n+1} + c$ where $n \in \mathbb{Q},\ n \neq -1$ and $a$ is a constant and $c$ is the **constant** of **integration**

Your notes

- Remember the special case:
  - $\int a \, \mathrm{d}x = ax + c$
    - e.g. $\int 4 \, \mathrm{d}x = 4x + c$
  - This allows **constant** terms to be integrated

## How do I integrate expressions containing powers of x?

- The formulae for integrating powers of $x$ apply to **all rational numbers** so it is possible to integrate any expression that is a sum or difference of powers of $x$
  - e.g. If $f(x) = 8x^3 - 2x + 4$ then
    $\int f(x) \, \mathrm{d}x = \frac{8x^{3+1}}{3+1} - \frac{2x^{1+1}}{1+1} + 4x + c = 2x^4 - x^2 + 4x + c$
- Functions involving **roots** will need to be rewritten as **fractional powers** of $x$ first
  - eg. If $f(x) = 5\sqrt[3]{x}$ then rewrite as $f(x) = 5x^{\frac{1}{3}}$ and integrate
- Functions involving **fractions** with **denominators** in **terms** of $x$ will need to be rewritten as **negative powers** of $x$ first
  - e.g. If $f(x) = \frac{4}{x^2} + x^2$ then rewrite as $f(x) = 4x^{-2} + x^2$ and integrate
- **Products** and **quotients** cannot be integrated this way so would need **expanding/simplifying** first
  - e.g. If $f(x) = 8x^2(2x - 3)$ then
    $\int f(x) \, \mathrm{d}x = \int (16x^3 - 24x^2) \, \mathrm{d}x = \frac{16x^4}{4} - \frac{24x^3}{3} + c = 4x^4 - 8x^3 + c$

### Examiner Tips and Tricks

- You can speed up the process of integration in the exam by committing the pattern of basic integration to memory
  - In general you can think of it as 'raising the power by one and dividing by the new power'

Your notes

- Practice this lots before your exam so that it comes quickly and naturally when doing more complicated integration questions

### Worked Example

Given that

$$\frac{dy}{dx} = 3x^4 - 2x^2 + 3 - \frac{1}{\sqrt{x}}$$

find an expression for $y$ in terms of $x$.

Rewrite all terms as powers of *x* using the laws of indices for fractional and negative powers on the last term.

$$\frac{dy}{dx} = 3x^4 - 2x^2 + 3 - x^{-\frac{1}{2}}$$

Find *y* by integrating each term.

$$y = \int \left(3x^4 - 2x^2 + 3 - x^{-\frac{1}{2}}\right) dx$$

$$y = \frac{3x^5}{5} - \frac{2x^3}{3} + 3x - \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + c$$

Rewrite using the same format given in the question.

$$\boldsymbol{y = \frac{3}{5}x^5 - \frac{2}{3}x^3 + 3x - 2\sqrt{x} + c}$$

# Finding the Constant of Integration

## How do I find the constant of integration?

- **STEP 1**
  - Rewrite the function into a more easily integrable form

    - Each term needs to be a power of **x** (or a constant)
- **STEP 2**
  - Integrate each term and remember "+c"
    - Increase power by 1 and divide by new power
- **STEP 3**
  - Substitute the coordinates of a given point in to form an equation in **c**
    - Solve the equation to find **c**

e.g. FIND THE EQUATION OF THE GRAPH THAT PASSES THROUGH (2,6) AND HAS GRADIENT FUNCTION $3x^2(2x-1)$

ANOTHER NAME FOR $\frac{dy}{dx}$

STEP 1 REWRITE IN A MORE EASILY INTEGRABLE FORM

$$\frac{dy}{dx} = 3x^2(2x-1)$$

$$= 6x^3 - 3x^2$$

STEP 2 INTEGRATE...

$$y = \frac{6x^4}{4} - \frac{3x^3}{3} + c$$

...REMEMBERING "+c"

$$y = \frac{3x^4}{2} - x^3 + c$$



Your notes

Your notes

STEP 3 $\boxed{\text{USE THE GIVEN POINT TO FIND } c}$

AT (2,6) $6 = \frac{3}{2}(2)^4 - 2^3 + c$

$c = -10$

$\therefore \quad y = \frac{3}{2}x^4 - x^3 - 10$



## Examiner Tips and Tricks

- If a constant of integration can be found then the question will need to give you some extra information
  - If this is given then make sure you use it to find the value of c

## Worked Example

Given $f'(x) = \dfrac{(x+3)^2}{\sqrt{x}}$ and $f(1) = 25$, find $f(x)$.

Rewrite $f'(x)$ in a form the can be integrated more easily.

Your notes

$$\mathrm{f}'(x) = \frac{(x+3)(x+3)}{\sqrt{x}}$$

$$= \frac{x^2 + 6x + 9}{x^{\frac{1}{2}}}$$

$$= \frac{x^2}{x^{\frac{1}{2}}} + \frac{6x}{x^{\frac{1}{2}}} + \frac{9}{x^{\frac{1}{2}}}$$

$$= x^{\frac{3}{2}} + 6x^{\frac{1}{2}} + 9x^{-\frac{1}{2}}$$

Integrate each term, remember to include a constant of integration.

$$\mathrm{f}(x) = \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + \frac{6x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{9x^{\frac{1}{2}}}{\frac{1}{2}} + c$$

Simplify.

$$\mathrm{f}(x) = \frac{2}{5}x^{\frac{5}{2}} + 4x^{\frac{3}{2}} + 18x^{\frac{1}{2}} + c$$

Use f(1) =25 to find the value of c.

$$\mathrm{f}(1) = \frac{2}{5}(1)^{\frac{5}{2}} + 4(1)^{\frac{3}{2}} + 18(1)^{\frac{1}{2}} + c$$

$$\frac{2}{5} + 4 + 18 + c = 25$$

$$22\frac{2}{5} + c = 25$$

$$c = \frac{13}{5}$$

$$f(x) = \frac{2}{5}x^{\frac{5}{2}} + 4x^{\frac{3}{2}} + 18x^{\frac{1}{2}} + \frac{13}{5}$$

Your notes

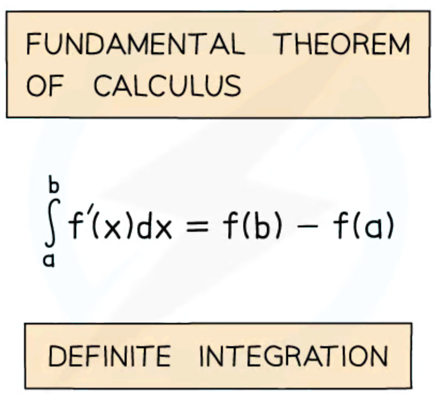
## Definite Integrals

Your notes

# Definite Integration

## What is definite integration?

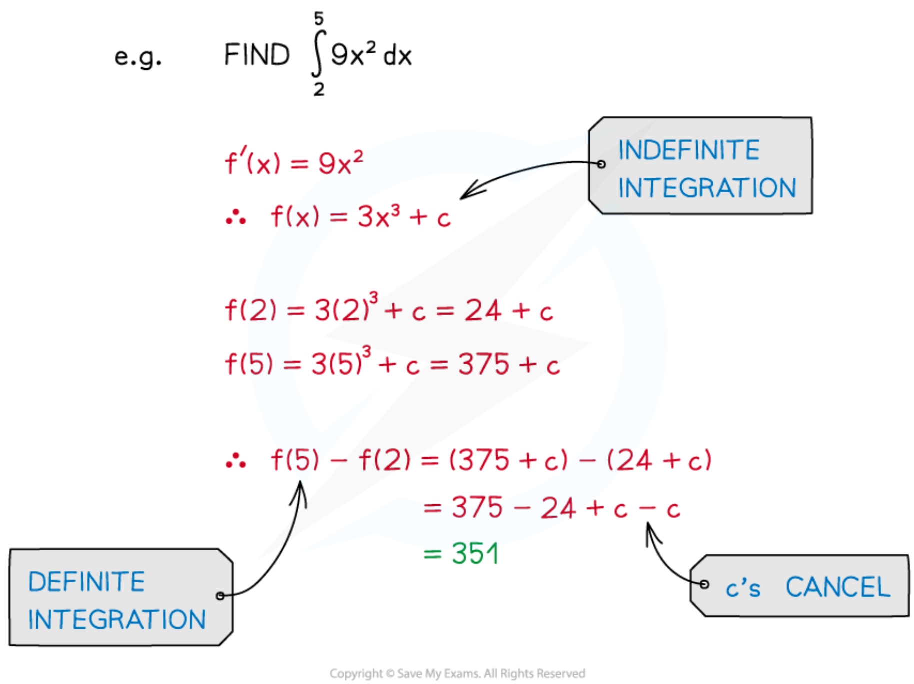
- **Definite Integration** occurs in an alternative version of the Fundamental Theorem of Calculus
- This version of the Theorem is the one referred to by most textbooks/websites






- **a** and **b** are called limits
  - **a** is the lower limit
  - **b** is the upper limit
- **f'(x)** is the **derivative** of **f(x)**
- The value can be **positive, zero** or **negative**

## Why do I not need to include a constant of integration for definite integration?




- "**+c**" would appear in both **f(a)** and **f(b)**
  - Since we then calculate **f(b)** – **f(a)** they cancel each other out
  - So "**+c**" is not included with definite integration

## How do I find a definite integral?

- **STEP 1**
  - Give the integral a name (if it does not already have one)
    - This saves you having to rewrite the whole integral every time!
- **STEP 2**
  - If necessary rewrite the integral into a more easily integrable form
    - Not all functions can be integrated directly
- **STEP 3**
  - Integrate without applying the limits
    - Notation: use square brackets [ ] with limits placed after the end bracket

Your notes

- **STEP 4**
  - Substitute the limits into the function and calculate the answer
    - Substitute the top limit first
    - Then substitute the bottom limit
    - Subtract the second value from the first

e.g. EVALUATE $\displaystyle\int_1^9 \left(2x^2 + \frac{3}{x^2} - 4\sqrt{x}\right)dx$

STEP 1 — NAME THE INTEGRAL, I

LET $\displaystyle I = \int_1^9 \left(2x^2 + \frac{3}{x^2} - 4\sqrt{x}\right)dx$

STEP 2 — REWRITE IN A MORE EASILY INTEGRABLE FORM

$$I = \int_1^9 \left(2x^2 + 3x^{-2} - 4x^{\frac{1}{2}}\right)dx$$

STEP 3 — INTEGRATE

- USE SQUARE BRACKETS
- LIMITS AT END

$$I = \left[\frac{2}{3}x^3 - 3x^{-1} - \frac{8}{3}x^{\frac{3}{2}}\right]_1^9$$

STEP 4 — SUBSTITUTE LIMITS AND CALCULATE

$$I = \left(486 - \frac{1}{3} - 72\right) - \left(\frac{2}{3} - 3 - \frac{8}{3}\right)$$

$$I = \frac{1256}{3}$$



Your notes

## What are the special properties of definite integrals?

Your notes

- Some of these have been encountered already and some may seem obvious ...
  - taking **constant** factors outside the integral
    - $\int_a^b k\mathrm{f}(x)\,\mathrm{d}x = k\int_a^b \mathrm{f}(x)\,\mathrm{d}x$ where $k$ is a constant
    - useful when fractional and/or negative values involved
  - integrating term by term
    - $\int_a^b [\mathrm{f}(x) + \mathrm{g}(x)]\,\mathrm{d}x = \int_a^b \mathrm{f}(x)\,\mathrm{d}x + \int_a^b \mathrm{g}(x)\,\mathrm{d}x$
    - the above works for subtraction of terms/functions too
  - equal upper and lower limits
    - $\int_a^a \mathrm{f}(x)\,dx = 0$
    - on evaluating, this would be a value subtracted from itself!
  - swapping limits gives the same, but negative, result
    - $\int_a^b \mathrm{f}(x)\,\mathrm{d}x = -\int_b^a \mathrm{f}(x)\,\mathrm{d}x$
    - compare 8 subtract 5 say, with 5 subtract 8 ...
  - splitting the interval
    - $\int_a^b \mathrm{f}(x)\,\mathrm{d}x = \int_a^c \mathrm{f}(x)\,\mathrm{d}x + \int_c^b \mathrm{f}(x)\,\mathrm{d}x$ where $a \le c \le b$
    - this is particularly useful for areas under multiple curves or areas under the $x$-axis

### Examiner Tips and Tricks

- Look out for questions that ask you to find an **indefinite** integral in one part (so "**+c**" needed), then in a later part use the same integral as a **definite** integral (where "**+c**" is not needed)

Your notes

## Worked Example

Your notes

? Find the exact value of $\int_0^8 (3x^{\frac{1}{2}} + \frac{4}{x^3} - \frac{\sqrt{x}}{x^2})\,dx$

STEPS 1 & 2 $I = \int_0^8 \left(3x^{\frac{1}{2}} + 4x^{-3} - x^{-\frac{3}{2}}\right)dx$

- CALL INTEGRAL I
- REWRITE

STEP 3 INTEGRATE

$$I = \left[\frac{3x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{4x^{-2}}{-2} - \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}}\right]_0^8$$

$$I = \left[2x^{\frac{3}{2}} - 2x^{-2} + 2x^{-\frac{1}{2}}\right]_0^8$$

STEP 4 SUBSTITUTE LIMITS AND CALCULATE

$$I = \left[2(8)^{\frac{3}{2}} - 2(8)^{-2} + 2(8)^{-\frac{1}{2}}\right] - \left[2(0)^{\frac{3}{2}} - 2(0)^{-2} + 2(0)^{-\frac{1}{2}}\right]$$

$$I = \left(32\sqrt{2} - \frac{1}{32} + \frac{\sqrt{2}}{2}\right) - (0 - 0 + 0)$$

$$I = \frac{-1 + 1040\sqrt{2}}{32}$$

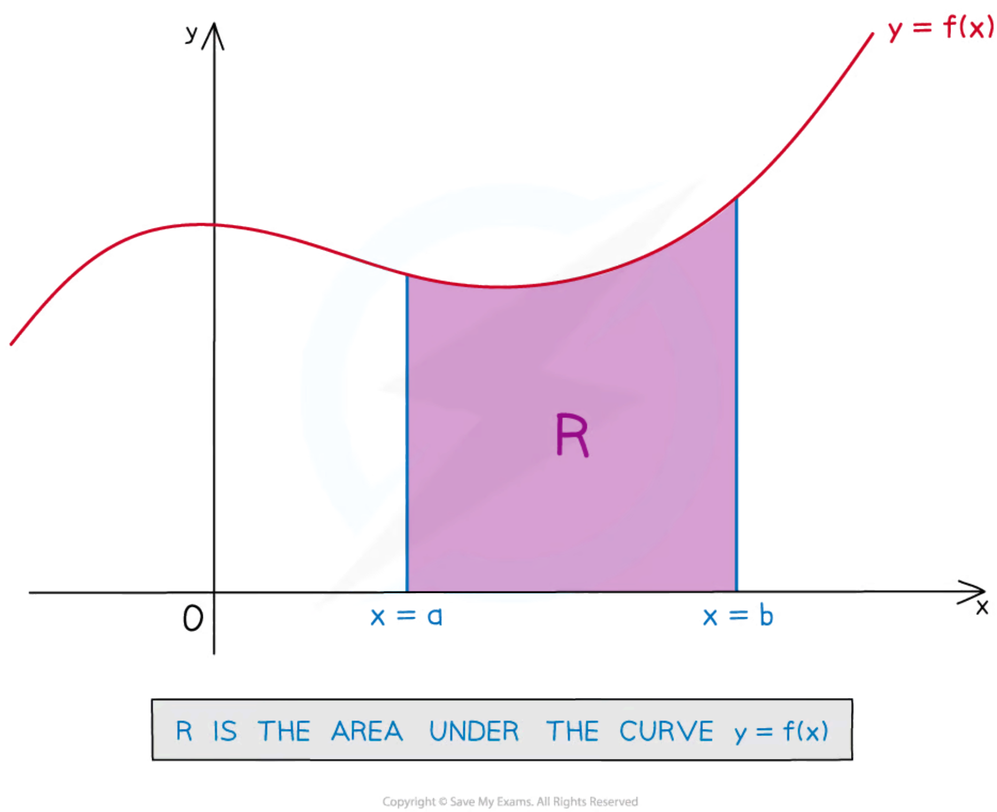
Your notes

## Finding Areas with Integration

# Area Under a Curve

## What does area under a curve mean?

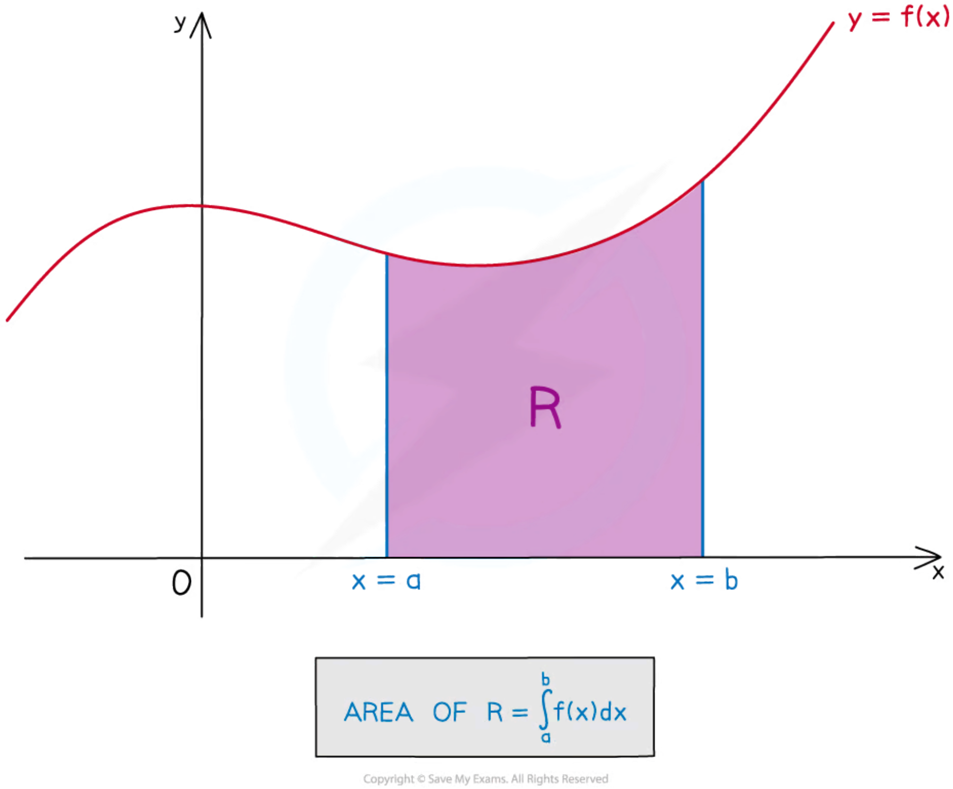
- The phrase "area under a curve" refers to the area bounded by ...
  - ... the **x**-axis
  - ... the graph of **y = f(x)**
  - ... the vertical line **x = a**
  - ... the vertical line **x = b**




## How do I find the area under a curve?

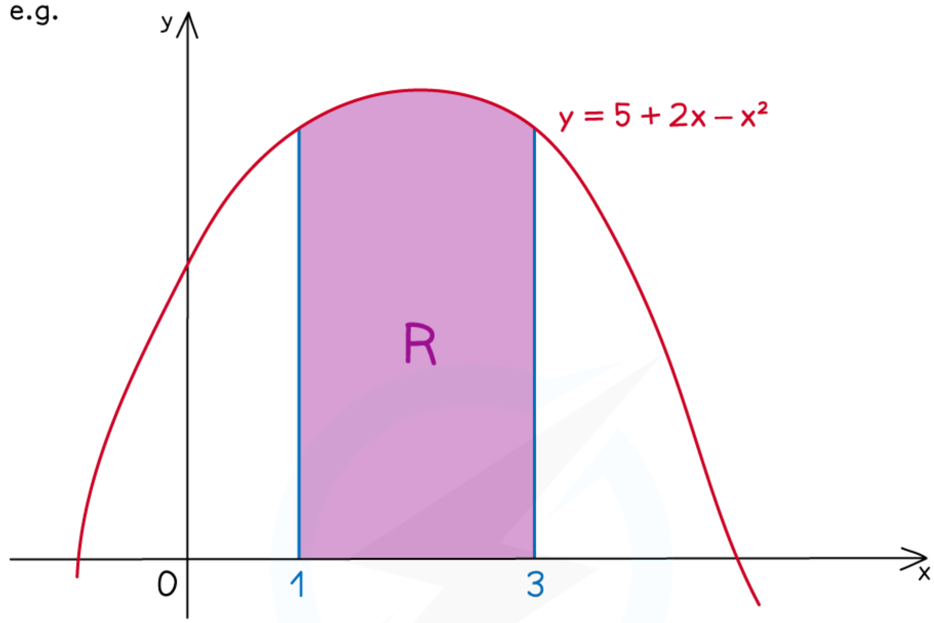
Your notes

- The value from **definite integration** is equal to the "area under a curve"



Your notes

e.g.




FIND THE AREA OF THE SHADED REGION.

$$\text{AREA} = \int_1^3 (5 + 2x - x^2)\,dx$$

$$= \left[5x + x^2 - \frac{1}{3}x^3\right]_1^3$$

$$= (15 + 9 - 9) - (5 + 1 - \frac{1}{3})$$

$$\text{AREA} = \frac{28}{3} \text{ SQUARE UNITS}$$

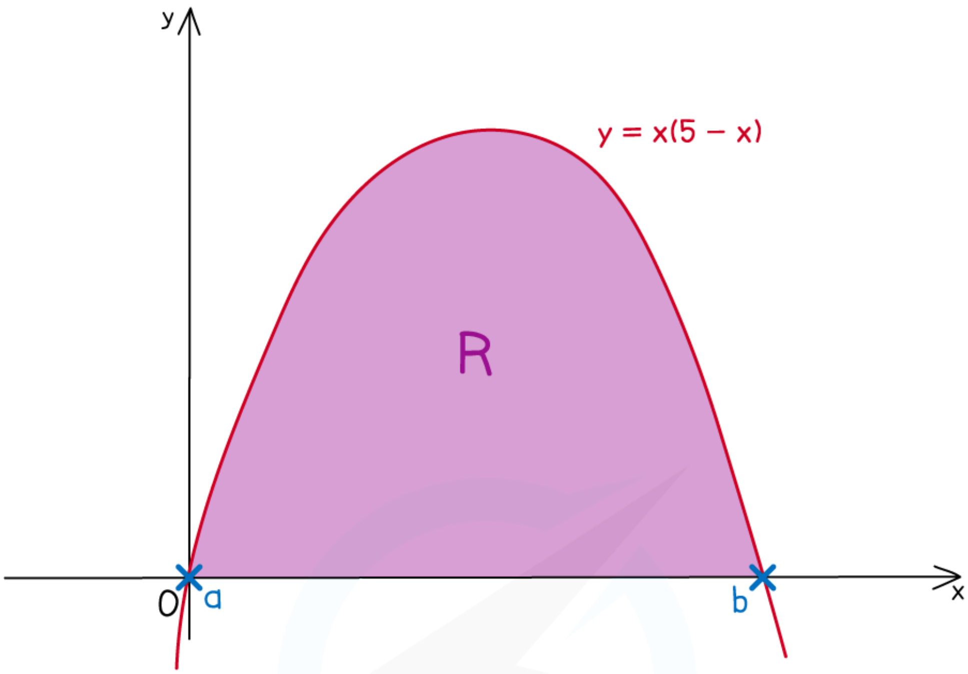


## What if I am not told the limits?

- If limits are not provided they will likely be the **x**-axis intercepts
  - Set **y = 0** and solve the equation to find the **x**-axis intercepts

Your notes

e.g. FIND THE SHADED AREA MARKED R IN THE DIAGRAM BELOW




$x(5 - x) = 0$ ← SET $y = 0$ FOR x–AXIS INTERCEPTS

$x = 0 \qquad x = 5$

$\therefore \quad a = 0 \qquad b = 5$

EXPAND SO "INTEGRATE–ABLE"

AREA OF R $= \int_0^5 (5x - x^2)\,dx$

$= \left[\frac{5x^2}{2} - \frac{x^3}{3}\right]_0^5$

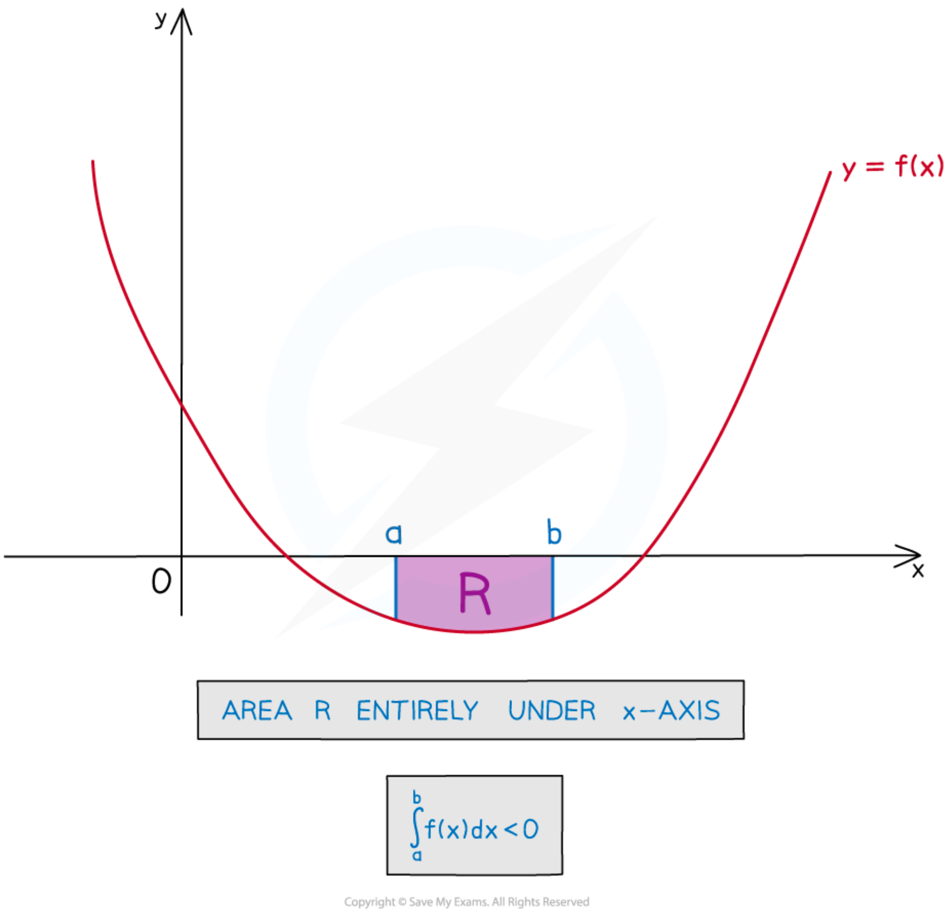
$= \frac{125}{2} - \frac{125}{3} - (0 - 0)$

AREA OF R $= \frac{125}{6}$ SQUARE UNITS

Your notes

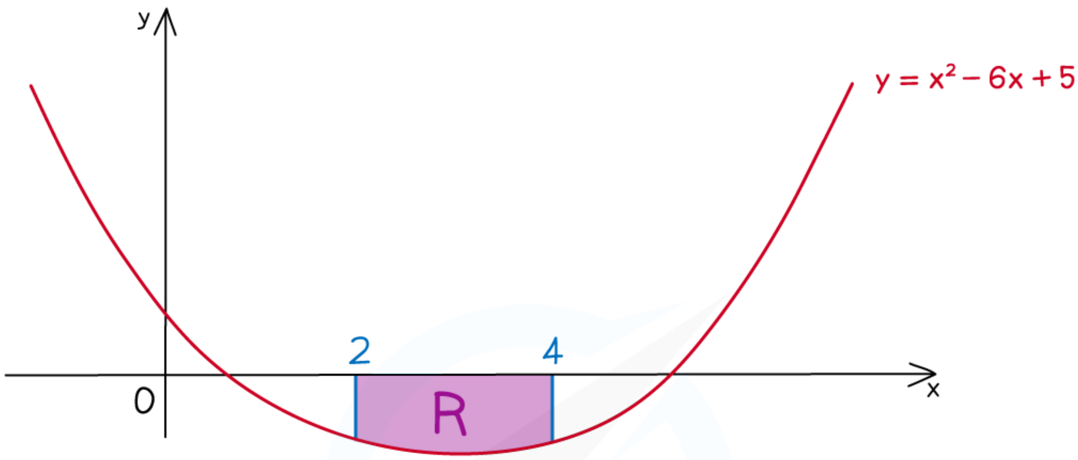
## What happens if the graph is below the x-axis?

- If the area lies underneath the **x**-axis the value of the integral will be negative
- An **area** cannot be negative, so take the **modulus** of the integral



Your notes

e.g. FIND THE SHADED AREA R IN THE DIAGRAM BELOW




$$\int_2^4 (x^2 - 6x + 5)\,dx = \left[\frac{x^3}{3} - 3x^2 + 5x\right]_2^4$$

WRITE AS AN INTEGRAL

$$= \left(\frac{64}{3} - 48 + 20\right) - \left(\frac{8}{3} - 12 + 10\right)$$

$$= -\frac{22}{3}$$

$\therefore$ AREA OF R $= \dfrac{22}{3}$ SQUARE UNITS

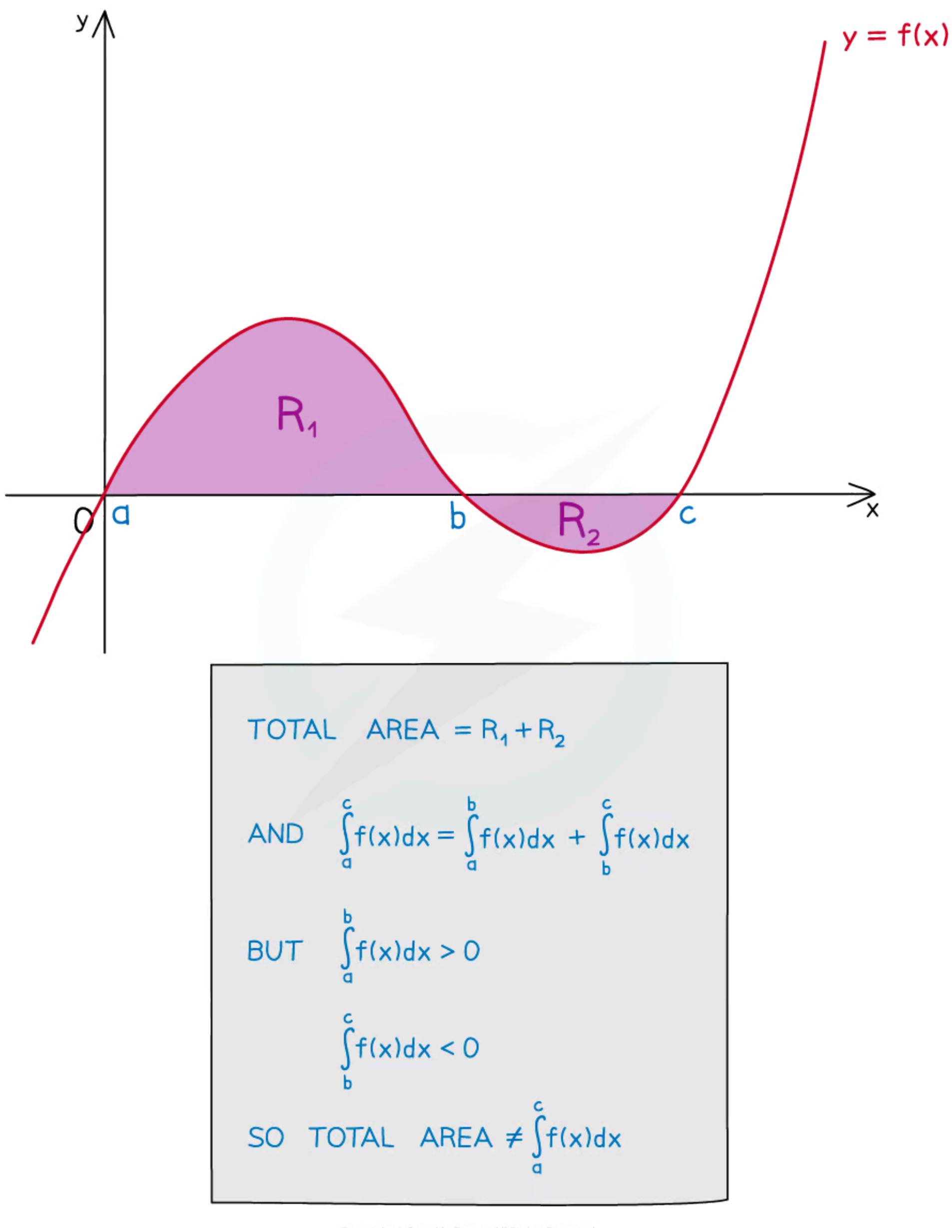
$$\left|-\frac{22}{3}\right| = \frac{22}{3}$$



## What if the area is made up of more than one section?

- Be careful when one section is above and one section is below the **x**-axis
  - You will need a separate integral for each section, BUT...
  - ...One section's integral will be **negative**
  - ...One section's integral will be **positive**

- So you'll need to take the **modulus** before adding to find the total area




Your notes

## Examiner Tips and Tricks

- Add information to any diagram provided in the question, as well as axes intercepts and values of limits
- Mark and shade the area you're trying to find, and if no diagram is provided, **sketch** one!

Your notes

## Worked Example

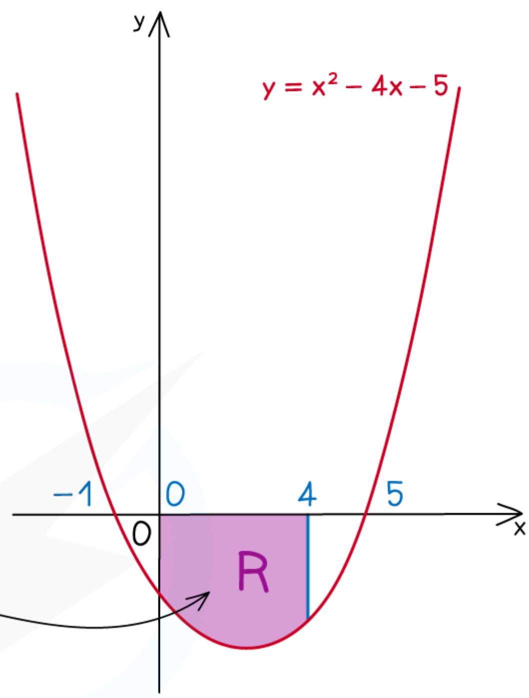
Your notes

Find the area bounded by the curve with equation $y = x^2 - 4x - 5$, the $x$-axis and the lines $x = 0$ and $x = 4$.

SKETCH AS NO DIAGRAM GIVEN

$x^2 - 4x - 5 = (x-5)(x+1) = 0$

$\therefore$ x-AXIS INTERCEPTS ARE $-1$ AND $5$




EXPECT NEGATIVE AS BELOW x-AXIS

$$\int_0^4 (x^2 - 4x - 5)\,dx = \left[\frac{x^3}{3} - 2x^2 - 5x\right]_0^4$$

WRITE AS AN INTEGRAL

$$= \left(\frac{64}{3} - 32 - 20\right) - (0 - 0 - 0)$$

$$= -\frac{92}{3}$$

$\therefore$ AREA REQUIRED $= \dfrac{92}{3}$ SQUARE UNITS

MODULUS

Your notes

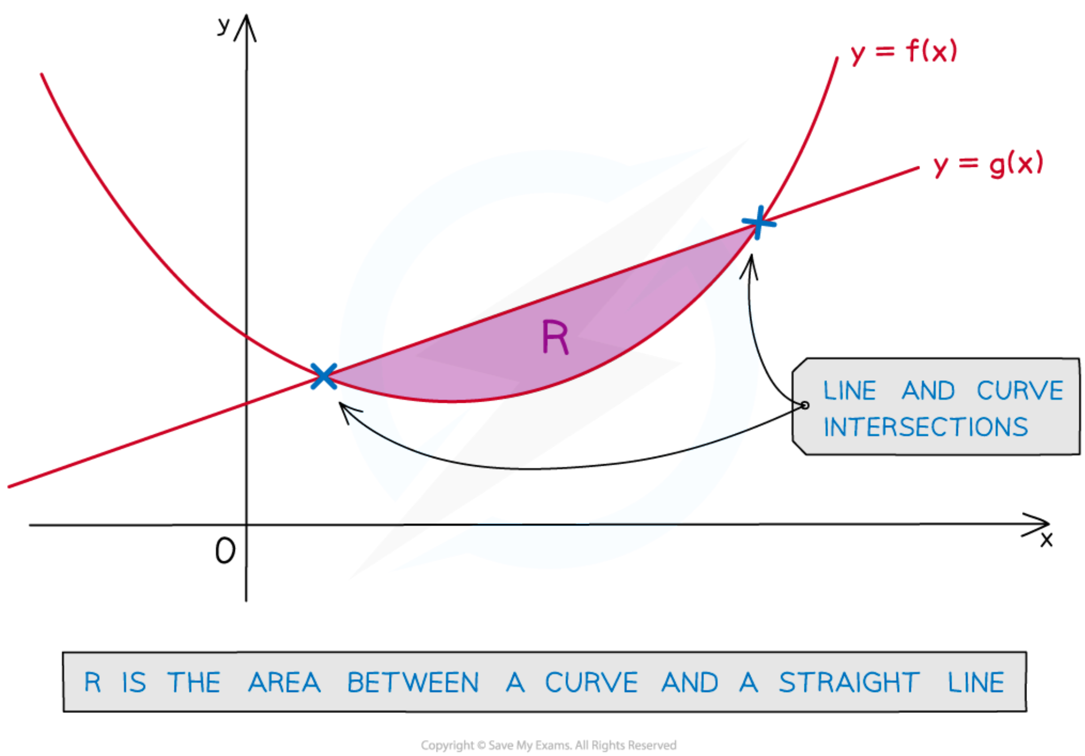
Finding Areas Between Lines & Curves

Your notes

# Area Between a Curve & a Line

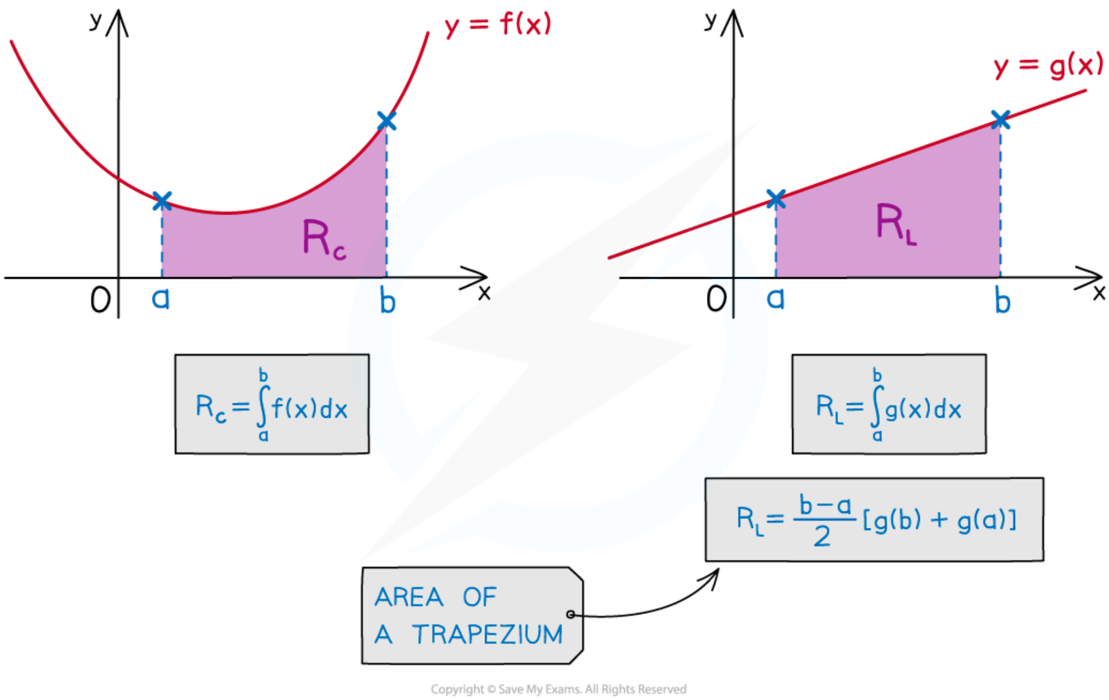
## How do I find the area between a curve and a line?

- The area enclosed will be the **difference** between ...
  - the **area under** the **line** and
  - the **area under** the **curve**




- These can be found separately

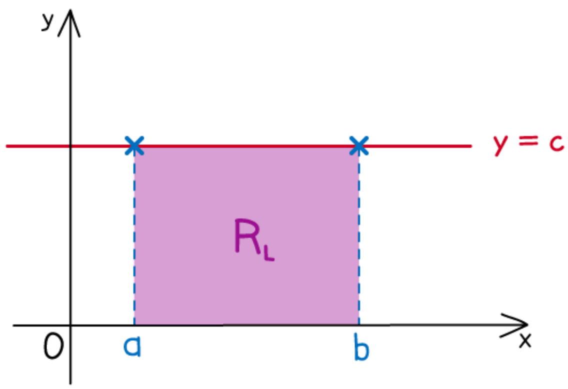
Your notes




- **STEP 1**
  - Find the intersections of the line and the curve (if not given)
- **STEP 2**
  - Find the area under a curve, $\mathbf{R_C}$, using definite integration
- **STEP 3**
  - Find the area under a line, $\mathbf{R_L}$, either using definite integration **or** the area formulae for basic shapes
- **STEP 4**
  - To find the area, **R**, between the curve and the line **subtract** the **smaller** area **from** the **larger** area
  - If curve on top this will be $\mathbf{R_C - R_L}$
  - If line on top this will be $\mathbf{R_L - R_C}$
- There may be easier ways to find the area under a line

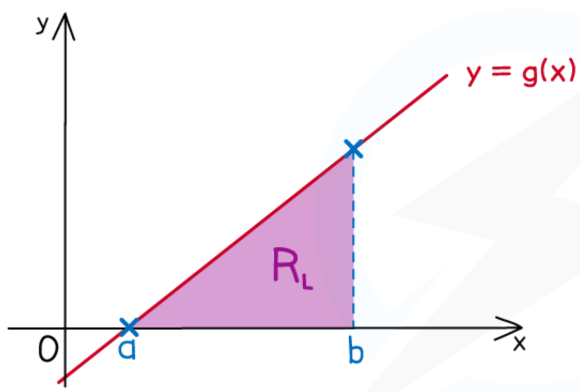
Your notes

AREA UNDER A LINE



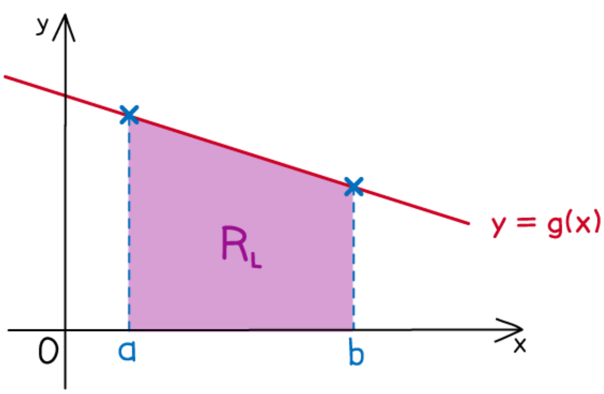

RECTANGLE OR SQUARE

$R_L = c(b - a)$




TRIANGLE

$R_L = \frac{1}{2}(b - a)g(b)$




TRAPEZIUM

$R_L = \frac{b-a}{2}[g(b) + g(a)]$

## Can I subtract before integrating?

- **Yes** - see the next note on "**Area between two curves**"
- Essentially you treat the line as a second curve
  - It can be easier to make mistakes using this method

### Examiner Tips and Tricks

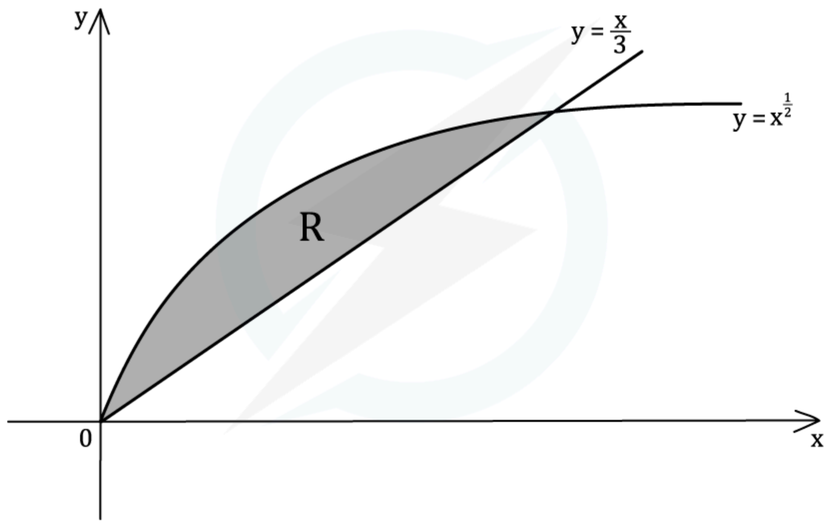
- Add information to any diagram provided
  - Add axes intercepts, as well as intercepts between lines and curves
- Mark and shade the area you're trying to find
- If no diagram is provided, **sketch** one!

### Worked Example

Your notes

Your notes

Find the area indicated by R in the diagram below.



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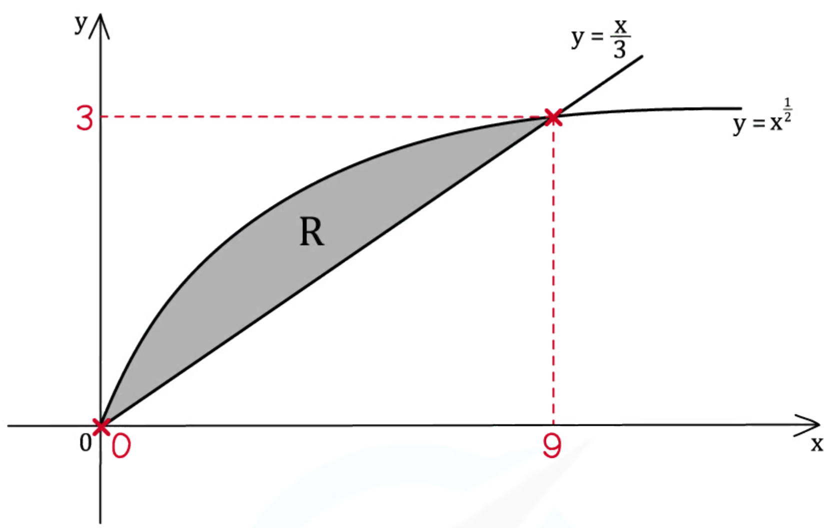



STEP 1 FIND THE INTERSECTIONS

$\frac{x}{3} = x^{\frac{1}{2}}$ ← SIMULTANEOUS EQUATIONS

$\frac{x^2}{9} = x$

$x^2 = 9x$ ← QUADRATIC

$x^2 - 9x = 0$

$x(x-9) = 0$

$x = 0$ AND $x = 9$ ← ADD TO DIAGRAM

Your notes

STEP 2 FIND THE AREA UNDER THE CURVE

$$R_c = \int_0^9 x^{\frac{1}{2}}\,dx = \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^9$$

$$= \left[\frac{2}{3}x^{\frac{3}{2}}\right]_0^9$$

$$= \left(\frac{2}{3}(9)^{\frac{3}{2}}\right) - (0)$$

$$R_c = 18$$

STEP 3 FIND THE AREA UNDER THE LINE

$$R_L = \frac{1}{2} \times 9 \times 3$$

TRIANGLE BASE = 9
HEIGHT = 3

$$R_L = \frac{27}{2}$$

Your notes

STEP 4 SUBTRACT SMALL FROM LARGE

$$R = R_c - R_L$$

CURVE IS ON TOP

$$R = 18 - \frac{27}{2}$$

$$R = \frac{9}{2} \text{ SQUARE UNITS}$$

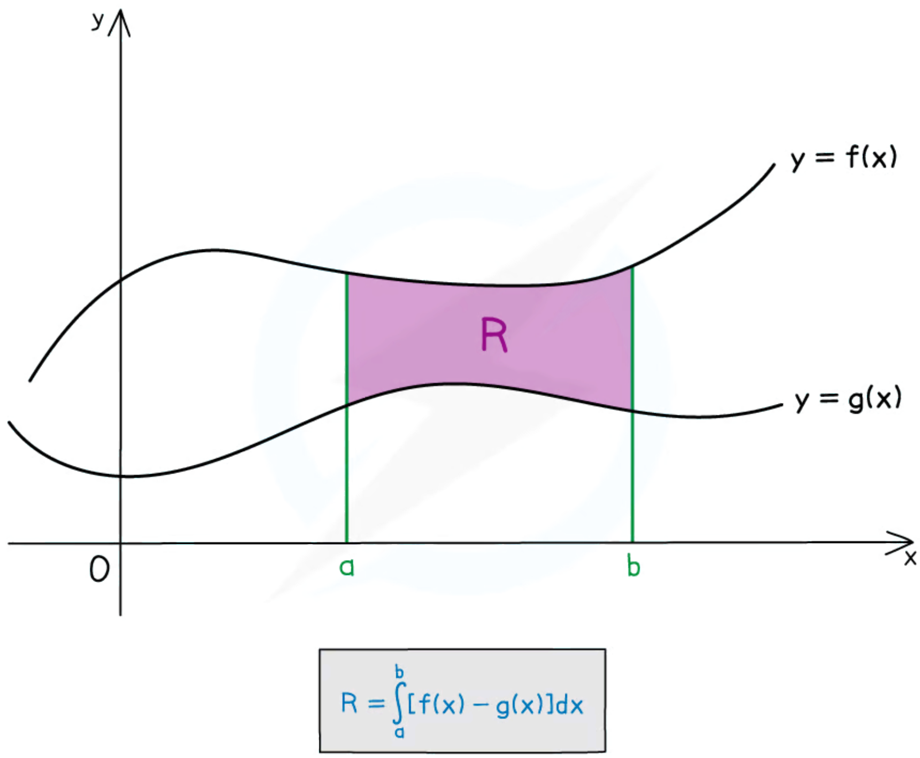


# Area Between Two Curves

## What is the area between two curves?

- In general find the definite integral of "upper curve" – "lower curve"
- However this does depend on ...
  - ... the area being found
  - ... if the curves intersect (and cross over)

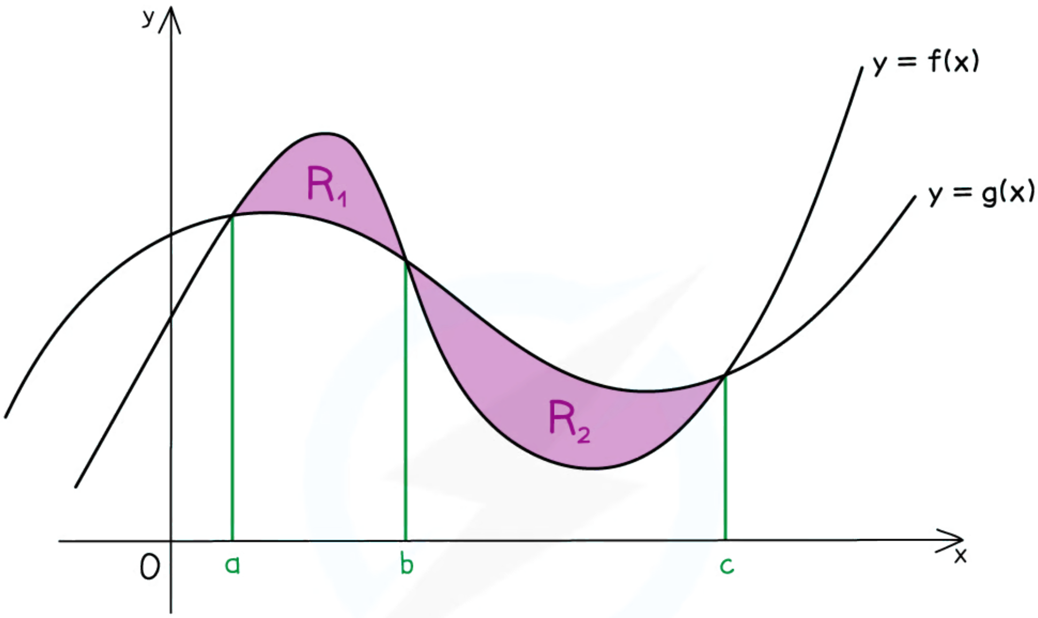
Your notes






- The area may have to be split into separate integrals

Your notes




$$R_1 = \int_a^b [f(x) - g(x)]dx$$

$$R_2 = \int_b^c [g(x) - f(x)]dx$$



- The points at which curves intersect may need to be calculated

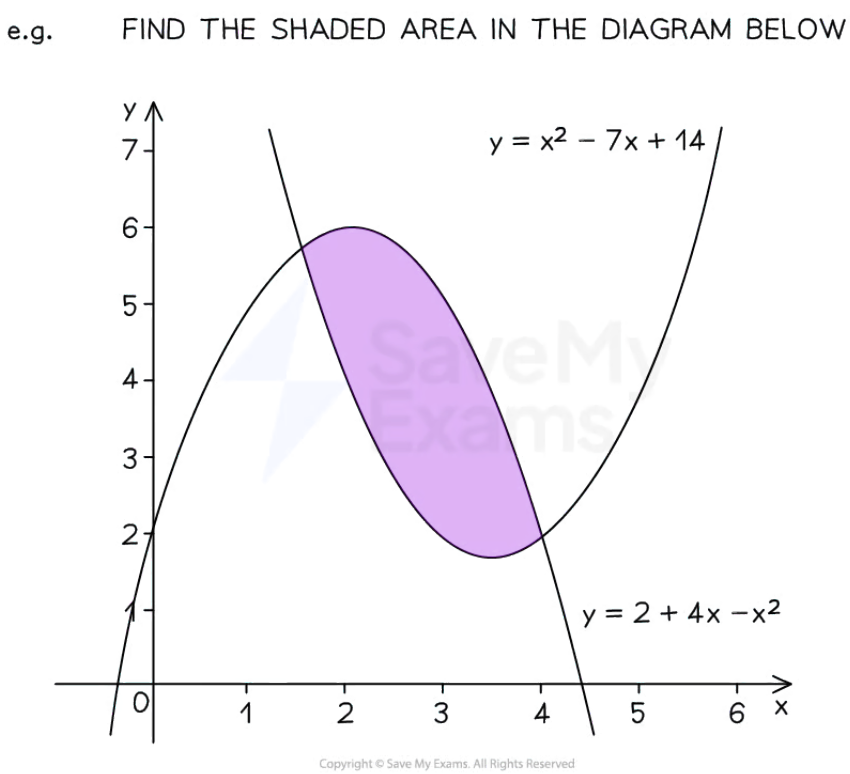
## How do I find the area between two curves?

- **STEP 1**
  - Find the intersections of the curves if needed
- **STEP 2**
  - Form the integral ...
    - ... using the intersections as limits
    - ... "upper curve" – "lower curve" ...

Your notes

- ... and find the value of the integral

- **STEP 3**
  - Repeat STEP 2 if more than one area needed
- **STEP 4**
  - Add areas together



Your notes

$\boxed{\text{STEP 1: FIND THE INTERSECTIONS}}$

$$x^2 - 7x + 14 = 2 + 4x - x^2$$
$$2x^2 - 11x + 12 = 0$$
$$(2x - 3)(x - 4) = 0$$
$$x = \frac{3}{2}, \qquad x = 4$$

$\boxed{\text{STEP 2: FORM INTEGRAL AND FIND ITS VALUE}}$

$$\text{AREA} = \int_{\frac{3}{2}}^{4} \left[\underbrace{(2 + 4x - x^2)}_{\text{UPPER CURVE}} - \underbrace{(x^2 - 7x + 14)}_{\text{LOWER CURVE}}\right] dx$$

$$\text{AREA} = \int_{\frac{3}{2}}^{4} (-2x^2 + 11x - 12)\, dx$$

$$\text{AREA} = \frac{125}{24} \text{ SQUARE UNITS}$$



### Examiner Tips and Tricks

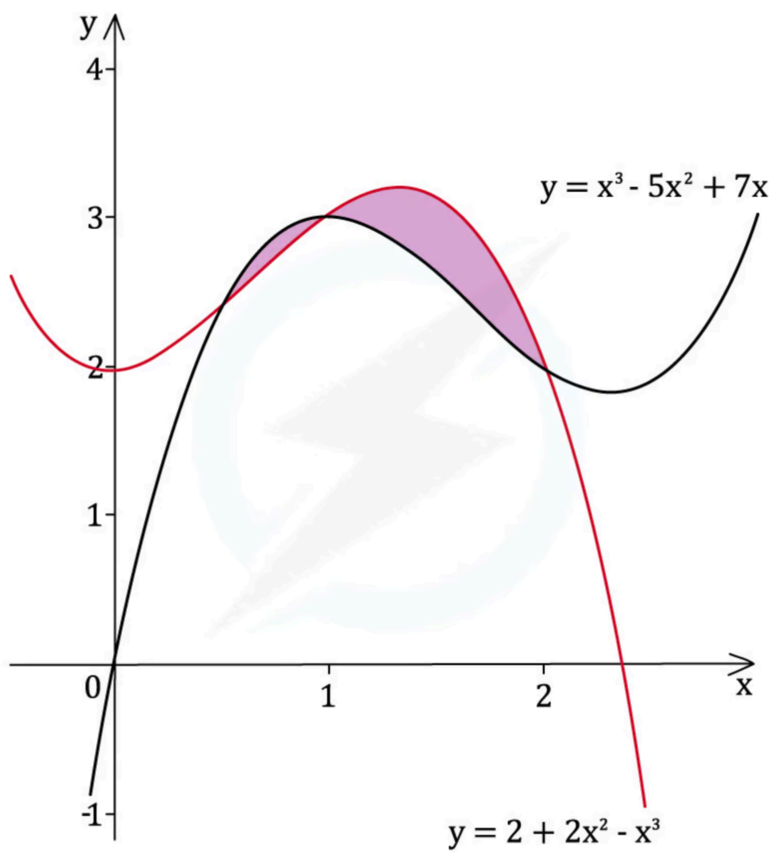
- If no diagram is provided sketch one, even if the curves are not accurate
- Add information to any given diagram as you work through a question

### Worked Example

Your notes

The diagram below shows a sketch of part of the graphs with equations

$y = x^3 - 5x^2 + 7x$ and $y = 2 + 2x^2 - x^3$




Find the shaded area.

Your notes

STEP 1: FIND THE INTERSECTIONS

$$x^3 - 5x^2 + 7x = 2 + 2x^2 - x^3$$

$$2x^3 - 7x^2 + 7x - 2 = 0$$

SOLVE USING CALCULATOR

$$x = 2, \quad x = 1, \quad x = \frac{1}{2}$$

STEP 2: FORM AND FIND INTEGRAL FOR FIRST AREA

$$R_1 = \int_{0.5}^{1} \left[ (x^3 - 5x^2 + 7x) - (2 + 2x^2 - x^3) \right] dx$$

$$R_1 = \int_{0.5}^{1} (2x^3 - 7x^2 + 7x - 2)\, dx$$

USE CALCULATOR

$$R_1 = \frac{5}{96}$$

STEP 3: FORM AND FIND INTEGRAL FOR SECOND AREA

$$R_2 = \int_{1}^{2} \left[ (2 + 2x^2 - x^3) - (x^3 - 5x^2 + 7x) \right] dx$$

$$R_2 = \int_{1}^{2} (-2x^3 + 7x^2 - 7x + 2)\, dx$$

$$R_2 = \frac{1}{3}$$

STEP 4: ADD AREAS TOGETHER

$$\text{SHADED AREAS} = R_1 + R_2$$

$$= \frac{5}{96} + \frac{1}{3}$$

$$\text{SHADED AREA} = \frac{37}{96} \text{ SQUARE UNITS}$$


Your notes

Your notes

## Integrating Trig Functions

# Integrating Trig Functions

## How do I integrate sin, cos and sec^2?

- The **integrals** for **sine** and **cosine** are

$$\int \sin x \, dx = -\cos x + c$$

$$\int \cos x \, dx = \sin x + c$$

where $c$ is the **constant** of **integration**

- Also, from the **derivative** of $\tan x$

$$\int \sec^2 x \, dx = \tan x + c$$

- For the **linear** function $ax + b$, where $a$ and $b$ are constants,

$$\int \sin(ax+b) \, dx = -\frac{1}{a}\cos(ax+b) + c$$

$$\int \cos(ax+b) \, dx = \frac{1}{a}\sin(ax+b) + c$$

$$\int \sec^2(ax+b) \, dx = \frac{1}{a}\tan(ax+b) + c$$

- For **calculus** with **trigonometric** functions **angles must** be **measured** in **radians**

### Examiner Tips and Tricks

- Remember to add 'c', the constant of integration, for any indefinite integrals

Your notes

## Worked Example

a) Find, in the form $\mathrm{f}(x) + c$, an expression for each integral

i. $\int \cos x \, \mathrm{d}x$

ii. $\int \sec^2\left(3x - \frac{\pi}{3}\right) \mathrm{d}x$

i. This is a result you should be able to recall from memory.

$$\int \cos x \, \mathrm{d}x = \mathbf{\sin x + c}$$

ii. Use the standard result: $\int \sec^2(ax+b) \, \mathrm{d}x = \frac{1}{a}\tan(ax+b) + c$.

$$\int \sec^2\left(3x - \frac{\pi}{3}\right) \mathrm{d}x = \mathbf{\frac{1}{3}\tan\left(3x - \frac{\pi}{3}\right) + c}$$

b) A curve has the following equation:

$$y = \int 2\sin\left(2x + \frac{\pi}{6}\right) \mathrm{d}x$$

The curve passes through the point with coordinates $\left(\frac{\pi}{3}, \sqrt{3}\right)$.

Find an expression for $y$.

Factor out the 2 in front of the sin first, and then use the standard result:

$$\int \sin(ax+b)\mathrm{d}x = -\frac{1}{a}\cos(ax+b) + c.$$

$$y = 2\int \sin\left(2x + \frac{\pi}{6}\right)\mathrm{d}x$$

$$y = 2\left(-\frac{1}{2}\cos\left(2x + \frac{\pi}{6}\right) + c\right)$$

Your notes

When multiplying the arbitrary constant $c$, by 2, it is still just an arbitrary constant and so it is still just written as $c$.

$$y = -\cos\left(2x + \frac{\pi}{6}\right) + c$$

Substitute in the given coordinate, and evaluate, to find $c$.

At $x = \frac{\pi}{3}$, $y = \sqrt{3}$

$$\sqrt{3} = -\cos\left(\frac{2\pi}{3} + \frac{\pi}{6}\right) + c$$

$$\sqrt{3} = -\cos\left(\frac{5\pi}{6}\right) + c$$

$$\sqrt{3} = \frac{\sqrt{3}}{2} + c$$

$$c = \frac{\sqrt{3}}{2}$$

Rewrite the full expression for $y$.

$$\boldsymbol{y = -\cos\left(2x + \frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}}$$

Integrating e^x & 1/x

Your notes

# Integrating e^x & 1/x

## How do I integrate exponentials and 1/x?

- The **integrals** involving $\mathrm{e}^x$ and $\ln x$ are

$$\int \mathrm{e}^x \, \mathrm{d}x = \mathrm{e}^x + c$$

$$\int \frac{1}{x} \, \mathrm{d}x = \ln|x| + c$$

where $c$ is the **constant** of **integration**

- For the **linear** function $(ax + b)$, where $a$ and $b$ are constants,

$$\int \mathrm{e}^{ax+b} \, \mathrm{d}x = \frac{1}{a}\mathrm{e}^{ax+b} + c$$

$$\int \frac{1}{ax+b} \, \mathrm{d}x = \frac{1}{a}\ln|ax+b| + c$$

- It follows from the last result that

$$\int \frac{a}{ax+b} \, \mathrm{d}x = \ln|ax+b| + c$$

- which can be deduced using **Reverse Chain Rule**
- With **ln,** it can be useful to write the constant of integration, $c$, as a logarithm
  - using the laws of logarithms, the answer can be written as a single term
  - $\int \frac{1}{x} \, \mathrm{d}x = \ln|x| + \ln k = \ln k|x|$ where $k$ is a constant
  - This is similar to the special case of **differentiating** $\ln(ax + b)$ when $b = 0$

Your notes

## Worked Example

A curve has the gradient function $f'(x) = \dfrac{3}{3x+2} + e^{4-x}$.

Given the exact value of $f(1)$ is $\ln 10 - e^3$ find an expression for $f(x)$.

To find $f(x)$, we need to integrate the expression for $f'(x)$.

$$f(x) = \int \frac{3}{3x+2} + e^{4-x}\,dx$$

Rewrite as two separate integrals which can be found individually, and factor out any constants.

$$f(x) = 3\int \frac{1}{3x+2}\,dx + \int e^{4-x}\,dx$$

Use the two results: $\displaystyle\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + c \quad \int e^{ax+b}dx = \frac{1}{a}e^{ax+b} + c$

$$f(x) = 3\left(\frac{1}{3}\ln|3x+2|\right) + -1e^{4-x} + c$$
$$f(x) = \ln|3x+2| - e^{4-x} + c$$

The question states that $f(1) = \ln 10 - e^3$, so substitute in $x = 1$ and equate to the given expression.

$$\ln 10 - e^3 = \ln|3(1)+2| - e^{4-1} + c$$

Simplify and solve to find $c$.

$$\ln 10 - e^3 = \ln 5 - e^3 + c$$
$$\ln 10 - \ln 5 = c$$

Recall from laws of logarithms that $\ln a - \ln b = \ln\dfrac{a}{b}$.

$$c = \ln\frac{10}{5} = \ln 2$$

Rewrite the full expression for $f(x)$.

$$f(x) = \ln|3x + 2| - e^{4-x} + \ln 2$$

or by combining logs, this could be written as $f(x) = \ln(2|3x + 2|) - e^{4-x}$

Your notes

Your notes

## Reverse Chain Rule

# Reverse Chain Rule

## What is the reverse chain rule?

- The **Chain Rule** is a way of differentiating two (or more) functions
- The **Reverse Chain Rule** (RCR) refers to **integrating** by **inspection**
  - Spotting that chain rule would be used in the reverse (differentiating) process

## How do I know when to use the reverse chain rule?

- The **reverse chain rule** is used when we have the **product** of a **composite function** and the **derivative** of its **second function**
- Integration is trickier than differentiation; many of the shortcuts do not work
  - For example, in general $\int \mathrm{e}^{\mathrm{f}(x)}\,\mathrm{d}x \neq \frac{1}{\mathrm{f}'(x)}\mathrm{e}^{\mathrm{f}(x)}$
  - However, this result **is true** if $\mathrm{f}(x)$ is linear $(ax + b)$
- Formally, in **function notation**, the **reverse chain rule** is used for **integrands** of the form

$$I = \int g'(x) f'(g(x))\,\mathrm{d}x$$

- This does not have to be strictly true, but '*algebraically*' it should be
- If the **coefficients** do not match '**adjust** and **compensate'** can be used
  - For example, $\mathrm{e}^{x^2}$ differentiates to $2x\mathrm{e}^{x^2}$ with the chain rule
    - so $\int 2x\mathrm{e}^{x^2}\,\mathrm{d}x = \mathrm{e}^{x^2} + c$ with the reverse chain rule
  - But to do $\int 5x\mathrm{e}^{x^2}\mathrm{d}x$ we need to:
    - **Take out** the five: $5\int x\mathrm{e}^{x^2}\mathrm{d}x$
    - Force a 2 inside (**adjust**) and divide the outside by a 2 (**compensate**): $\frac{5}{2}\int 2x\mathrm{e}^{x^2}\,\mathrm{d}x$

Your notes

- The bit inside the integral is now a reverse chain rule
- The answer is $\frac{5}{2}e^{x^2} + c$
- A particularly useful instance of the reverse chain rule to recognise is

$$I = \int \frac{f'(x)}{f(x)}\, dx = \ln|f(x)| + c$$

- i.e. the **numerator** is (almost) the **derivative** of the **denominator**
  - '**adjust** and **compensate**' may need to be used to deal with any coefficients
    - e.g.

$$I = \int \frac{x^2+1}{x^3+3x}\, dx = \frac{1}{3}\int 3\frac{x^2+1}{x^3+3x}\, dx = \frac{1}{3}\int \frac{3x^2+3}{x^3+3x}\, dx = \frac{1}{3}\ln|x^3+3x| + c$$

## Examiner Tips and Tricks

- You can always check your work by differentiating, if you have time

## Worked Example

A curve has the gradient function $f'(x) = 5x^2\sin(2x^3)$.

Given that the curve passes through the point $(0,\ 1)$, find an expression for $\mathrm{f}(x)$.

Write f($x$) as an integral.

$$\mathrm{f}(x) = \int 5x^2 \sin(2x^3)\, dx$$

Take 5 out of the integral as a factor.

$$\mathrm{f}(x) = 5\int x^2 \sin(2x^3)\, dx$$

The main function is sin(...), which would have come from -cos(...).

Adjust and compensate the coefficients. $2x^3$ would differentiate to $6x^2$ so -cos($2x^3$) would differentiate to ($6x^2$)sin($2x^3$)

Your notes

$$f(x) = 5 \times \left(\frac{1}{6}\right) \times \int (6x^2)\sin(2x^3)\,dx$$

Integrate.

$$f(x) = \left(\frac{5}{6}\right) \times -\cos(2x^3) + c$$

$$\mathbf{f(x) = -\frac{5}{6}\cos(2x^3) + c}$$

# Integrating Composite Functions (ax+b)

## What is a composite function?

- A **composite function** involves one function being applied after another
- A composite function may be described as a "function of a function"
- This Revision Note focuses on one of the functions being **linear** – i.e. of the form $\boldsymbol{ax + b}$

## How do I integrate linear (ax+b) functions?

- The reverse chain rule can be used for integrating functions in the form $y = (ax + b)^n$
  - Make sure you are confident using **the chain rule** to **differentiate** functions in the form $y = (ax + b)^n$
  - The reverse chain rule works backwards
- For $n = 2$ you will most likely **expand the brackets** and **integrate each term separately**
- If $n > 2$ this becomes **time-consuming** and if $n$ is not a positive integer we need a different method completely
- To use the **reverse chain rule** $\int (ax + b)^n\,dx$ (provided $n$ is not -1)
  - **Raise** the **power of *n* by 1**
  - **Divide** by this **new power**
  - **Divide** this whole function by the **coefficient of *x***
    - $\int (ax + b)^n\,dx = \dfrac{(ax + b)^{n+1}}{n+1} \times \dfrac{1}{a} + c$
- You can **check your answer by differentiating it**

Your notes

  - You should get the original function when you differentiate your answer
- Note that this method **only works** when the function in the brackets is **linear (*ax* + *b*)**
- The special cases for **trigonometric functions** and **exponential** and **logarithmic functions** are
  - $\int \sin(ax+b)\,\mathrm{d}x = -\frac{1}{a}\cos(ax+b) + c$
  - $\int \cos(ax+b)\,\mathrm{d}x = \frac{1}{a}\sin(ax+b) + c$
  - $\int \mathrm{e}^{ax+b}\,\mathrm{d}x = \frac{1}{a}\mathrm{e}^{ax+b} + c$
  - $\int \frac{1}{ax+b}\,\mathrm{d}x = \frac{1}{a}\ln|ax+b| + c$
- $c$, in all cases, is the **constant** of **integration**
- All the above can be deduced using **reverse chain rule**
  - However, spotting them can make solutions more efficient

## Worked Example

Find the following integrals

a) $\int 3(7-2x)^{\frac{5}{3}}\,\mathrm{d}x$

Name the integral.

$$I = \int 3(7-2x)^{\frac{5}{3}}\mathrm{d}x = 3\int (7-2x)^{\frac{5}{3}}\mathrm{d}x$$

Using the rule 'raise the power by one, divide by the new power and then multiply by the reciprocal of the derivative' integrate the expression.

Your notes

$$I = 3\left[\frac{1}{\frac{8}{3}}(7-2x)^{\frac{8}{3}} \times \frac{1}{-2}\right] + c$$

Simplify.

$$I = 3\left[-\frac{3}{16}(7-2x)^{\frac{8}{3}}\right] + c$$

$$\boldsymbol{I = -\frac{9}{16}(7 - 2x)^{\frac{8}{3}} + c}$$

b) $\int \frac{1}{2}\cos(3x-2)\,\mathrm{d}x$

Name the integral.

$$I = \int \frac{1}{2}\cos(3x-2)\mathrm{d}x = \frac{1}{2}\int \cos(3x-2)\mathrm{d}x$$

Using the rule $\int \cos(ax+b)\,\mathrm{d}x = \frac{1}{a}\sin(ax+b) + c$, integrate the expression.

$$I = \frac{1}{2}\left[\frac{1}{3}\sin(3x-2)\right] + c$$

Simplify.

$$\boldsymbol{I = \frac{1}{6}\sin(3x-2) + c}$$